



August 2022

The Dreaded “D” Word

Drawdowns!!!! Probably the worst word in a trader’s vocabulary.

Drawdown is a word that enlists all kinds of emotional feelings and intellectual thoughts – all negative.

But here’s the thing: Drawdowns are inevitable for career traders. It does not matter how you trade; you will sooner or later (mostly sooner) experience a drawdown. And drawdowns occur with regularity and usually at the most inappropriate times.

It’s how a trader understands and reacts to drawdowns that determines his/her future as a trader. Make no doubt about it, how a trader handles adversity makes a greater difference than how a trader handles victorious periods.

Misunderstand or mistreat drawdowns, and you will have a very limited lifespan as an investor/market speculator.

All career traders have a unique perspective on drawdowns. I can only share my own experiences and viewpoints on the subject, including my observations of dozens of other traders.

So, here goes.

What is a DD?

A drawdown is a measured decline in the value of an investment/trading account due to a sequence of net losing trades, including changes in the value of open trades, followed by a new equity high peak.

Drawdowns can be measured in two primary ways:

- Depth – % decline in the nominal account value from a peak to a valley prior to an eventual new NAV peak
- Duration – length of time from a peak account value to the depth of a DD to a new NAV peak

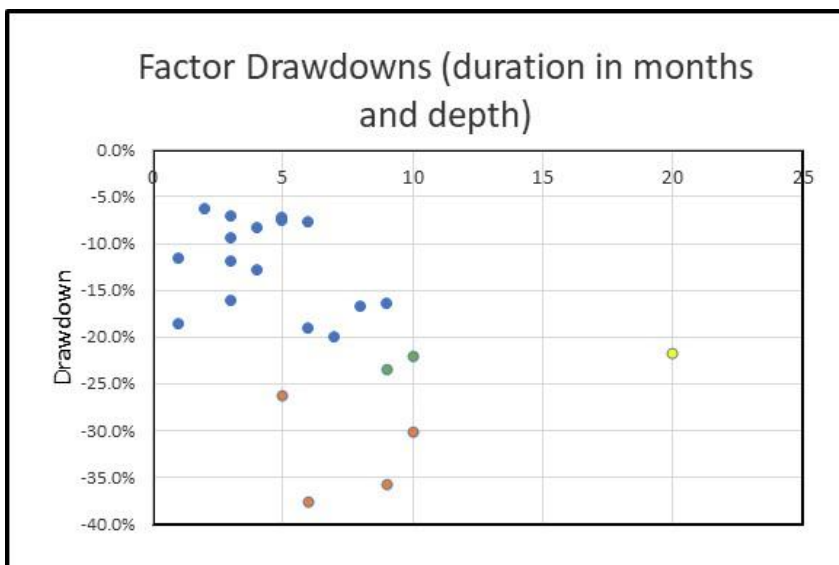
Commodity trading advisors (CTAs) and hedge funds measure DDs on end-of-month valuations as required by U.S. regulators. Some professional traders measure their DDs on an end-of-week basis for their own use.

I am aware that many retail traders monitor their account value on a daily or even intraday basis, although I personally think that such a short-term obsession with the value of a trading account can produce irrational trading decisions. IMO it is unhealthy to constantly monitor the value of a trading account.

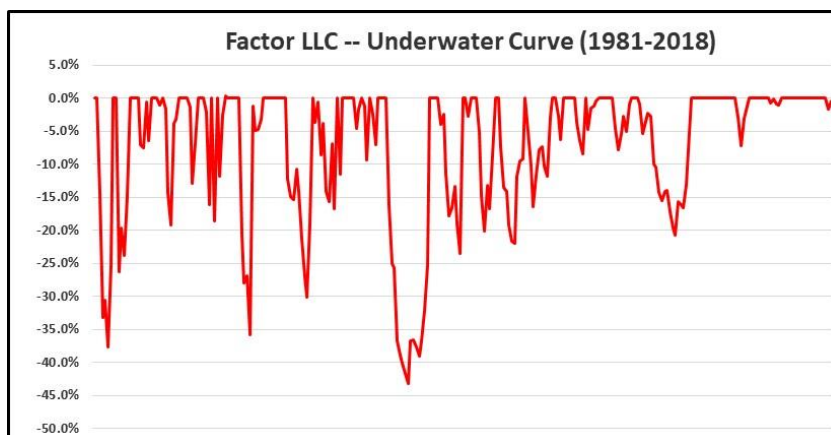
Prior to 2014 I tracked my proprietary trading performance according to CFTC standards – month-ending basis, marked-to-the market. I have always arbitrarily considered a drawdown to be a month-end to month-end decline in the Net Asset Value (NAV) of my notional proprietary accounts greater than 5%. Other traders use different criteria to determine a DD. Because professional traders frequently deposit and withdraw funds from accounts, the value of accounts should be tracked on a NAV basis with time-weighted adjustments for additions/withdrawals, not as absolute USD values. The percent gain or loss for any measurement period (usually monthly) is factored into the NAV from the previous period producing the Value-Added Monthly Index (VAMI). VAMI is really the same thing as a cume NAV.

As shown on the graph right, I have endured 20 month-to-month-to month DDs greater than 5% during my career, three greater than 30% and six longer than nine months, with the longest being 20 months. The 20-month DD occurred in 2013–2014 and included my worst annual loss at (13.7%).

Remember, these DDs were tracked on a month-ending basis. I am sure my performance has had dozens and dozens of 5%, even >10% DDs if measured on a daily basis.



Another representation of DD used by CTAs is the Under Water Curve (Factor Prop Account shown). This depicts the periods that a trader is either in a DD or recovering from a DD. In reality, most traders live mostly in a DD or recovering from a DD. If this seems strange, think about Bitcoin. Bitcoin has been underwater since Nov 8, 2021 and was underwater from Dec 2017 until Dec 16, 2020, a period of 36 months.



In 2014 for reasons beyond the scope of this paper, I began tracking performance in an entirely new way – as a closed trade NAV which adds/subtracts the percent gain/loss of each sequential closed trade to a running NAV. For tax purposes my accountant reports to the IRS my trading on a marked-to-the-market basis (includes open trades).

Why do drawdowns occur?

Following I discuss two types of Drawdowns:

- Drawdown ending in destruction or disappointment
- Drawdowns within long-term profitability

Drawdown ending in destruction or disappointment

Here is an important point: Technically the concept of "Drawdowns" applies to long-term profitable trading programs because eventual new NAV highs are implied. A trader whose account zig-zags in a downward slope over time or never really gains traction technically does not experience Drawdown per se.

Over the years brokerage houses, regulatory agencies (such as NFA and CFTC) and academic researchers have consistently reported that fewer than 20% of futures and forex traders are profitable after three years. After five years the proportion increases because of survivors' bias. My guess is that the data for crypto and equity traders are very different. Crypto traders who began prior to 2021 would have much higher success profiles – crypto traders who began since 2021 would be similar to futures and FX traders. Stock traders who are long-term holders have also done well. Short-term stock traders would fall into line with futures/FX traders.

There are many reasons why a trader might never gain traction. I will highlight several.

Absence of a trading plan. As insane as it sounds, many people who engage in market speculation do so without a conceived organized plan or goal. This applies to the process of identifying a candidate trade, the sizing of trades, overall risk management and managing individual trades.

Note: There are a very small proportion of traders who do quite well, even from the start, based on instinct and feel rather than rules, but these individuals are a huge exception. How does a person know if he or she is in this group? Answer: Profitability comes naturally and continually. I am NOT in this category.

Non-commitment to a trading program (lack of conviction). A trader may have most or all the pieces for a successful trading program, including:

- Insightful market analysis
- Trade identification
- Entry process
- Risk and trade management guidelines/rules
- Organized procedures
- Performance reviews
- Process for adopting modifications to plan
- Safeguards to protect plan from emotional swings

But still, traders may have lacked the conviction to go "all-in" following the plan. Self-doubt and second-guessing are huge enemies of trading success. A trader who does not follow his or her trading plan does not really have a trading plan.

Serial seekers of the holy grail. Many novice/aspiring traders search for the holy grail, which of course has never existed. These traders look for the easy answers. They jump from trading approach to trading approach, trading service to trading service and training program to training program to find the end of the proverbial rainbow. But due to self-doubt they never stick with one approach long enough.

Bet sizing destined to fail. A trader who risks 10% or more of their capital on trade bets is guaranteed to quickly encounter a drawdown and then to eventually destroy the account beyond redemption. Early in my career I

placed huge bets on some trades. To this day I almost start shaking when I think about these trades – it is just plain luck that I survived such foolishness. Even 5% bets will most likely end badly.

Inability to endure drawdowns. An aspiring trader might have developed or be following an approach that is very sound, but the trader bails out during a severe drawdown, giving up often at the very low of the drawdown. This can also lead to a “pick-and-choose” trading style, which also usually ends badly.

Trading plan destined for failure. The reality is that some trading plans, even when perfectly executed, do not have an edge, meaning that they will eventually erode trading capital. The obvious problem is that a trader may not realize the lack of an edge until most of the capital investment has been depleted. This is why I highly encourage the use of paper trading accounts for a period of two to three years. Of course, there are some inherent problems with paper trading, but I will leave this for another day.

Drawdowns within long-term profitability

This paper will now address drawdowns within the context of a long-term profitable trading plan. Some reasons for drawdowns within a profitable plan are easy to comprehend, another is rather complex.

Trading plan out of synch with markets. No trading plan is perfect. For example, trend following systems do great when markets are trending, but tend to lose money in choppy sideways markets. Mean-reversion trading can be wonderful in highly volatile markets, but miserable in calm sideways markets.

Here is the performance table of Dunn Capital Management, a trend following CTA that has been extremely profitable since 1984. I'd have no hesitation to place capital with Dunn, especially after the firm has experienced a losing year or two. Dunn has remained committed to the same basic trading plan for many years, of course with periodic refinements.

NET MONTHLY RETURNS (%)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YEAR
2022	9.14	8.45	13.52	8.41	-1.24								43.87
2021	-1.66	3.38	1.64	7.66	2.54	-4.50	-2.57	-0.71	1.13	8.78	-15.30	6.02	4.12
2020	0.17	-5.77	7.79	0.43	-2.87	-3.47	-1.20	1.67	-5.22	-0.22	-0.43	8.01	-2.10
2019	-3.44	1.59	10.88	1.57	0.99	4.28	4.89	12.10	-11.16	-6.04	2.86	-0.47	16.91
2018	10.72	-19.66	-0.38	5.80	-6.78	1.38	-1.54	3.71	2.63	-11.96	-5.85	3.96	-19.97
2017	-0.51	5.87	-0.94	-0.29	-0.74	-3.14	-2.54	-0.48	0.23	14.11	3.38	6.11	21.70

Trader out of synch with trading plan. This condition applies to discretionary traders. Due to many reasons a discretionary trader can get out of synch with his or her trading program. The reasons could be various, such as getting sloppy or lazy, second-guessing signals, periodic FOMO, emotional stress due to personal reasons, etc. Skilled traders can experience some of the same flaws of novice traders, yet find a way to get back into gear with good results.

Experimentation gone wrong. There is the human tendency for traders to modify their systematic or discretionary approach in the direction of those aspects of the trading plan that have recently worked best. This most often proves to be a mistake. This is especially true during DD periods.

Back in 2013 I entered a DD. At the time there was a lot of talk among traders that the nature of price behavior had changed and that technical tools that worked in the past were no longer viable. I bought into this lie and

began experimenting with new tools and indicators. The outcome was a DD that was two to three times longer than had I not made any changes to my approach.

Only when I returned to the basics of classical charting with aggressive risk management in 2014 was my trading able to recover. All approaches to trading have good periods and bad periods. While minor evolutionary fine-tuning over time is necessary, traders need to be VERY hesitant to modify an approach with a profitable history.

The unexpected effect of randomized outcomes. During my extended 2013–2014 drawdown I became extremely interested to determine effect of random probability on trading performance. In other words, to what degree are DDs of varying depths and durations the product of statistical probability?

Think of it this way:

- Let's say my win rate is 50%. I may absolutely believe my next trade will be profitable, yet there is an equal chance it will be unprofitable
- I might believe five of my next 10 trades will be profitable, yet it is within one standard deviation that seven of the next 10 trades will be losers. Among the seven might be my worst loser of the year
- There is also a statistical probability that two of the next 10 trades will be at my maximum loss level (60 BPs) and that six or seven of the winners will be marginal
- In other words, random probability can cause some strange outcomes

Over the course of three years, let's assume I have 396 trades. I know certain things (metrics) about these trades:

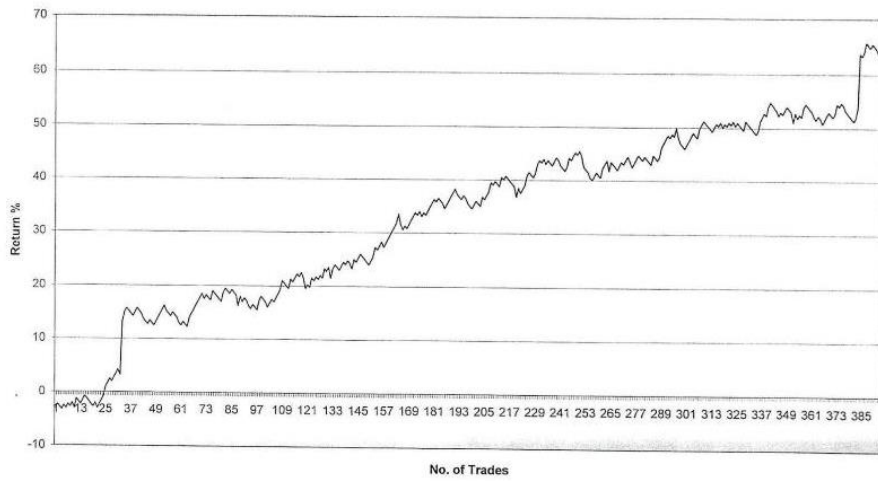
- % Profitable and unprofitable (i.e., win rate)
- Largest loss
- Average gain
- % Bet size
- % of trades residing into varying levels of gain and loss sizes

But the metrics during some sequence of 20 or 50 or even 100 of these trades can be quite different.

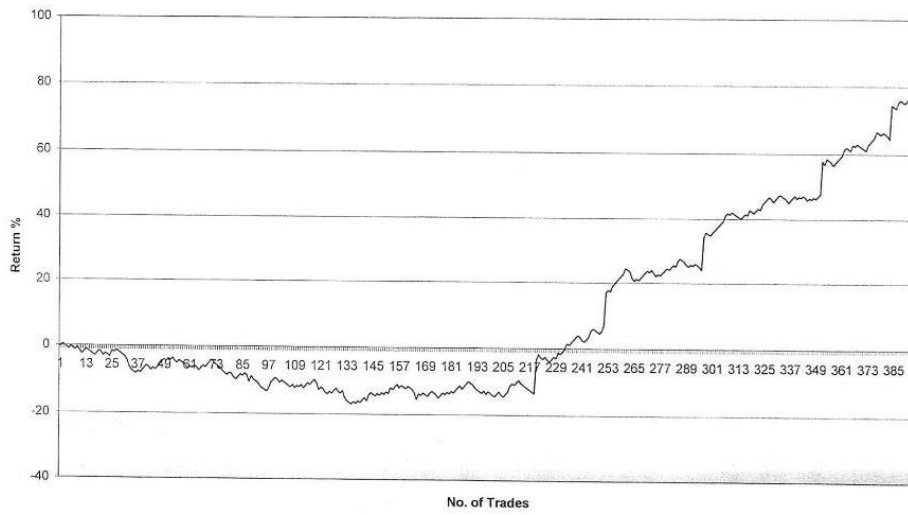
Next, I run these metrics through an Excel randomizer over thousands of trades. This is like having a deck of 3,960 cards, the total of which match the metrics of the 396 trades. By shuffling this deck of cards and plotting the equity curve, one card at a time, the outcome of each successive 396 cards can be quite different.

An actual example can be found in the graphs following. The prop trading group at Goldman London constantly test systematic trading systems. They then run the metrics of the trading systems through a randomizer to get an idea of the different equity curves the SAME EXACT metrics might produce within one and two standard deviations. Graphs of possible equity curves from the same trading metrics can be found on the next two pages.

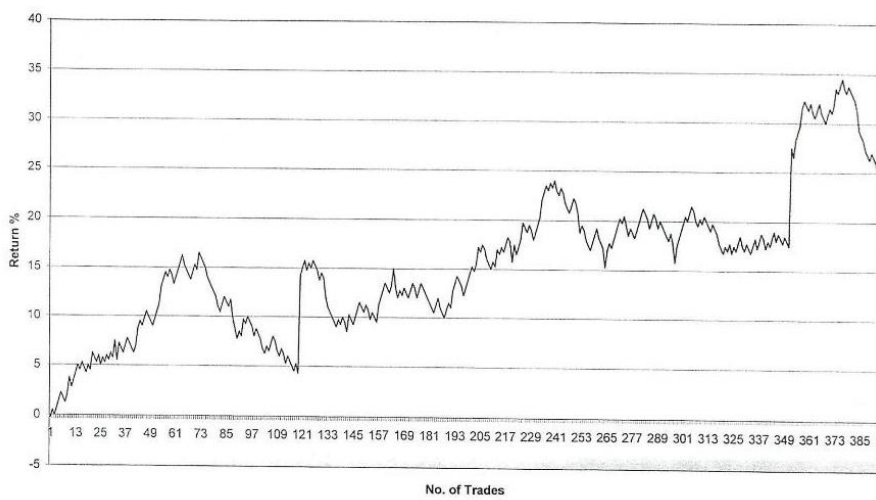
396 Trades (3 Years)

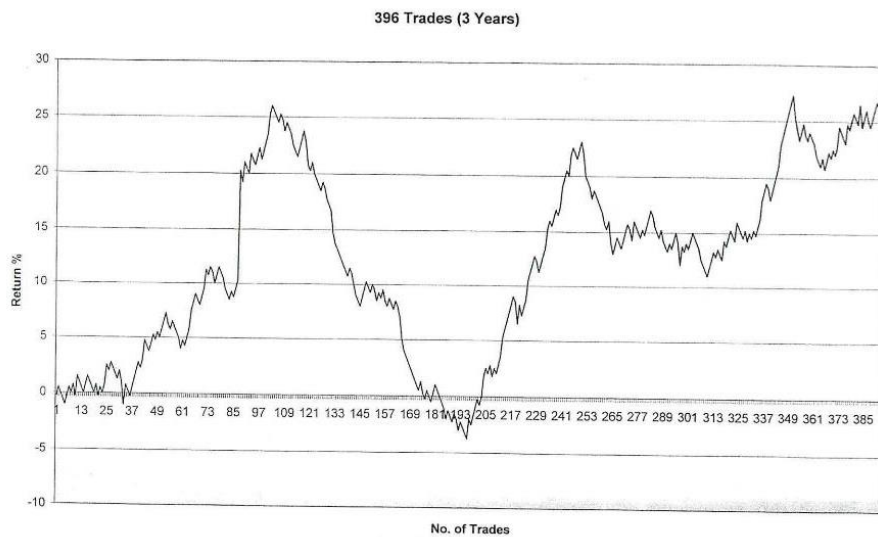
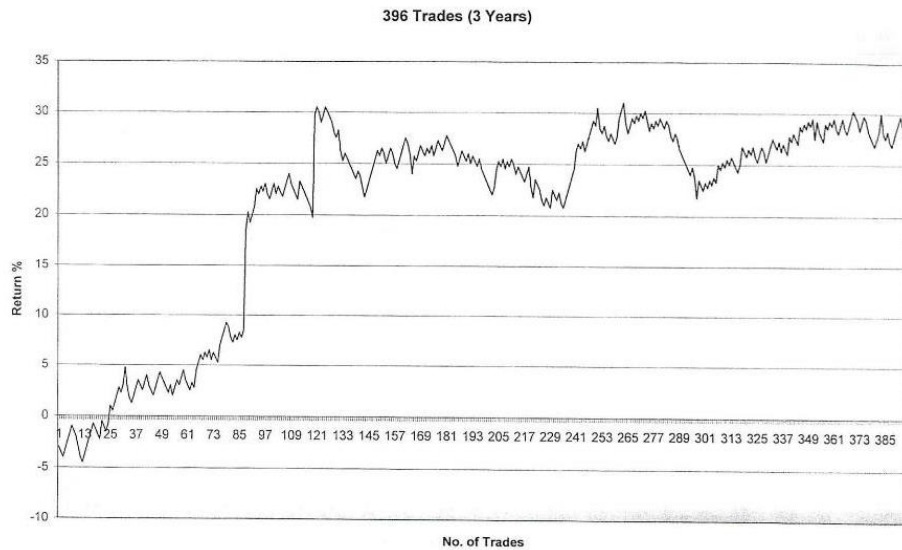


396 Trades (3 Years)



396 Trades (3 Years)





As you can see, random probability can be a real bitch. Some strange things can occur on route to long-term profitability. There are places along the timeline even quite profitable trading programs when traders ask questions such as:

- *"Will I ever have a winning trade again?"*
- *"Does my trading program even work?"*
- *"Should I stop trading for period?"*

Random probability promises times of uncertainty in all trading programs.

Next steps

For everyone who wishes for longer-term success in day trading/scalp trading/swing trading/systematic trading there comes a time of reconning when the following steps must be taken:

Formulate personalized methods to:

- Manage overall risk
- Evaluate markets for trades
- Generate trading signals
- Manage ongoing trades
- Develop processes for order entry
- Combat emotional influences

Then:

- Go all in – take a leap of faith
- Start with smaller bet sizes until success is verified
- Commit to disciplined execution for three years
- Have a monetary stop-loss (suggest 30%)
- Formulate performance evaluation measures

The Leap of Faith is the absolute hardest part of the equation, not knowing if the inevitable DDs are part of the process or signs that your trading plan does not have an edge (destined to fail).

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